Available online at www.sciencedirect.com
SCIENCE DIRECT®
Acta Tropica 89 (2004) 147–159
ACTA TROPICA
www.elsevier.com/locate/actatropica

Malaria transmission in relation to rice cultivation in the irrigated Sahel of Mali

Guimogo Dolo [a], Olivier J.T. Briët [b,*], Adama Dao [a], Sékou F. Traoré [a], Madama Bouaré [a], Nafomon Sogoba [a], Oumou Niaré [a], Magaran Bagayogo [a], Djibril Sangaré [a], Thomas Teuscher [b], Yeya T. Touré [a]

[a] *Département d'Epidémiologie des Affections Parasitaires, Faculté de Médecine, de Pharmacie et d'Odonto-Stomatologie, BP 1805, Bamako, Mali*
[b] *West Africa Rice Development Association, 01 BP 2551 Bouaké 01, Bouaké, Côte d'Ivoire, Ivory Coast*

Abstract

Seven cross-sectional entomological surveys were carried out from September 1995 to February 1998 in three irrigated rice growing villages and three villages without irrigated agriculture in the area surrounding Niono, located 350 km north-east of Bamako, Mali. The transmission pattern differed markedly between the two zones. In the irrigated zone, the transmission of malaria was fairly constant over the seasons at a low level. In the non-irrigated zone, transmission was mostly below detection level during the dry season, whereas it was high toward the end of the rainy season. In the irrigated zone, high densities of mosquitoes were correlated with low anthropophily, low sporozoite indices and probably low survival rates. In the non-irrigated zone, mosquito densities were lower and these relationships were less pronounced. Differential use of mosquito nets in the two zones may have been an important factor in the observed differences in transmission. The presence of cattle may also have played an important role. Two mosquito-catching methods (human landing catch and spray catch) were compared.
© 2003 Elsevier B.V. All rights reserved.

Keywords: *Anopheles gambiae*; *Anopheles funestus*; Mosquitoes; Rice; Irrigation; Malaria; Transmission; Mali; West Africa

1. Introduction

In the Sahel, the short rainy season of three to 4 months restricts mosquito breeding. The introduction of irrigation schemes and construction of dams for hydroelectric power generation have provided semi-permanent water surfaces, which could serve as breeding sites. Rice fields have proved to be particularly well suited as breeding sites for *Anopheles gambiae s.l.*, the main malaria vector in sub-Saharan Africa. This heliophilic species thrives in the shallow inundated fields during tilling, transplanting, the first 6 weeks of the growing period (until canopy closure), and after harvest (Klinkenberg et al., 2003). The Office du Niger is one of the biggest and oldest irrigation schemes in West Africa, with more than 50 000 ha of arable land, which is mainly used for rice cultivation. In this irrigation scheme, only part of the fields can be cultivated in the dry season due to limited water

* Corresponding author. Present address: IWMI, P.O. Box 2075, Colombo, Sri Lanka. Tel.: +94-112-787-404; fax: +94-112-786-854.
E-mail address: o.briet@cgiar.org (O.J.T. Briët).

0001-706X/$ – see front matter © 2003 Elsevier B.V. All rights reserved.
doi:10.1016/j.actatropica.2003.10.014

resources. The rotation of the start of irrigation of the fields and the existence of fields cultivated with one rice crop per year simultaneously with fields with two crops, with overlapping growing seasons, provide almost continuous breeding sites. This causes large mosquito populations in the irrigation scheme as compared to the surrounding Sahel, particularly during the dry season. This research studied whether the extended breeding season and increased mosquito populations caused by irrigated rice cultivation have effects on malaria transmission, as compared to the non-irrigated Sahel.

2. Material and methods

2.1. Study area

The Office du Niger (Fig. 1) was created in 1932 around the town of Niono (14°17′N, 8°5′W), with the objective of starting an agricultural revolution, producing cotton and rice for francophone West Africa. Water is drawn from the Niger River. The cultivation of cotton has been abandoned after problems with pest pressure and rising of the water table. Rehabilitation projects intervened in the 1980s and high-input

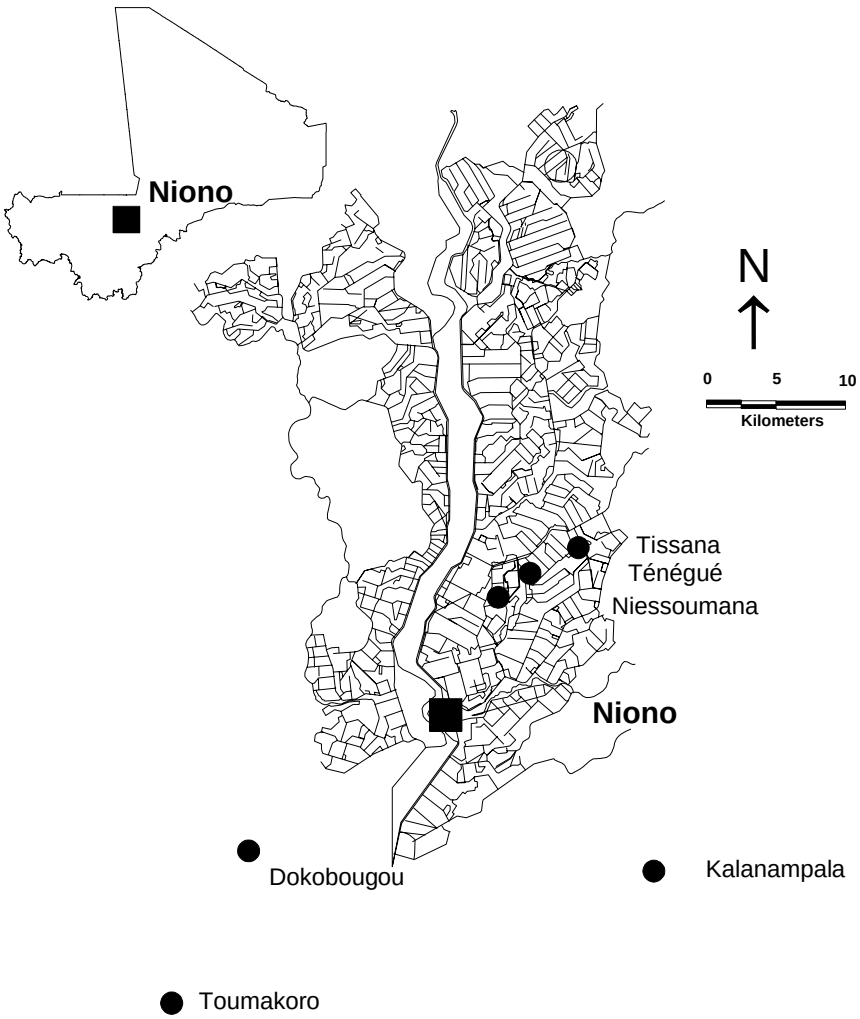


Fig. 1. Map showing the study villages and the irrigation canals of the Office du Niger around the city of Niono. The position of Niono in Mali is indicated in the insert.

technologies such as transplanting technique, modern varieties and high dosages of fertiliser were introduced. Since 1990, total liberalisation of the rice market, introduction of double cropping and the introduction of privately owned husking machines increased yields to an average of 5.5 t/ha (1998). The climate is typical of the Sahel, with a rainy season of about 4 months (July to September/October) characterised by heavy rains from the south, a 'cold' dry season characterised by the harmattan winds from the north (November to February), and hot dry season from March to May/June. Besides rice, cultivation of vegetables, rearing of cattle and fishing are practised.

In the irrigated zone, the population is estimated at 150 000 people, and the major ethnic groups are the Miniaka, Bambara, Peulh, Bela, and Bozo. Islam is the most important religion, but there are also a few Christians and animists. The zone is a vast plain of sandy clay with numerous dikes, primary, secondary, and tertiary irrigation canals and drains. Fields surround the villages, which are spread out over the scheme, to the visible horizon. The vegetation consists of crops, aquatic plants that grow alongside and inside the canals, and a few trees. Houses are rectangular and made of *banco* (clay bricks), with *terasse* (earth) or corrugated iron roofs.

In the area surrounding the irrigation scheme (the non-irrigated zone), the vegetation is open savannah of the Sahel type with thorny shrubs and few large trees (baobab and balazan). The major ethnic groups are the Bambara, Peulhs, Maures, and Sarakolés. Mainly millet and sorghum are cultivated by the Bambara. Cattle, sheep and goats are reared by the Peulh. Houses are made of banco, those of the Bambara are rectangular with earthen roofs; those of the Peulh are round with thatched roofs. Domestic animals live inside the courts.

After a preliminary study in September 1995, six typical villages—three from the irrigated zone that practice double rice cropping (Ténégué, Tissana, Niessoumana) and three from the non-irrigated zone with mainly millet cultivation (Dokobougou, Toumakoro, Kalanampala)—were selected on criteria of accessibility and willingness of villagers to co-operate. For a further description of the villages in the irrigated zone; see Sissoko et al. (2004).

2.2. Mosquito collection

Two visits were made in the cold dry season: in January 1997 and February 1998; two visits were made when the off-season rice crop was cultivated in the irrigated zone: in April 1996 and April 1997; and three visits were made in the middle and end of the rainy season: in August 1995, August/September 1996 and October 1997. During each visit, two methods of mosquito collection were undertaken: human landing catch (night catch) and spray catch.

2.3. Human landing catch (night catch)

The human landing catch was performed from 18:00 to 06:00 h inside and outside of human dwellings by volunteers recruited in the villages. Each visit, two night catches at two collection points per session were performed, thus a total of eight man-nights per village per survey. Mosquitoes were collected in 2 h intervals. Anopheline mosquitoes were identified, counted and dissected to determine the parity ratio using Detinova's method (Detinova, 1962). Each individual anopheline mosquito was divided into two parts (abdomen and head−thorax) and kept frozen at −20 °C in labelled vials with Carnoy's fixative. The abdomens were used to identify the species (*A. gambiae s.s.* versus *A. arabiensis*) by PCR. The head–thorax parts were used to determine the Circum Sporozoite Protein (CSP) infection rate or sporozoite index (the proportion of female anopheline mosquitoes carrying sporozoites in the head−thorax) by ELISA. For large catches, a random sample of mosquitoes was taken, the size of the sample being in proportion to the size of the catch in each 2 h interval.

2.4. Spray catch

Two pyrethrum-spray catch sessions were performed per visit in each village from 15:00 to 17:00 h in human dwellings. Up to 45 (average 32) sleeping rooms (depending on the expected mosquito density) were sprayed per village located in the irrigated zone, and up to 60 (average 57) rooms were sprayed per village located in the non-irrigated zone, recording at the same time the number of sleepers per house. Anopheline mosquitoes were identified, classified by abdominal status (unfed, blood fed, semi-gravid,

gravid), counted and processed. All mosquitoes were stored as described for the mosquitoes in the human landing catch. The abdomens (ovaries) of the semi-gravid females of *A. gambiae s.l.* were used for chromosomal identification. The remainder of the blood meal of the semi-gravid mosquitoes and the blood meals of the blood-fed mosquitoes were analysed by ELISA for the human blood index (the proportion of the fed mosquitoes that fed on humans). The abdomens of the unfed, fed and gravid *A. gambiae s.l.* mosquitoes were processed for species identification by PCR. The head−thorax parts of all classes of the mosquitoes were processed by ELISA to determine the CSP infection rate.

2.5. Data analysis

The data were recorded on standard forms and entered in a database editor (dBase version 5). Data were analysed using the packages SPSS 9.0 and MS Excel 97. The feeding success was determined as the proportion of blood-fed and semi-gravid mosquitoes in the total population presumed to have been trying to feed (all mosquitoes except gravids), following Charlwood et al. (1995). We did not correct for partially fed mosquitoes. The feeding success for each species was compared between zones with the Mann–Whitney *U*-test and Mantel–Haenszel χ^2-test. With the method of Lines et al. (1991a) we investigated whether the ratio between the two methods of the estimation of aggressiveness of the mosquito population (number of mosquitoes per person per night from the night catch and the spray catch) changed with density. If there was no correlation between the ratio and density, linear regression was applied. If the ratio changed with density, linear regression was applied with log-transformed variables. The relationships between the different entomological parameters were analysed with correlation and regression analysis. The entomological inoculation rates (EIRs) of *Anopheles gambiae s.l.* and *Anopheles funestus* combined, from the spray catch (blood fed and semi-gravid females per person per night multiplied by the Human Blood Index and CSP Index) were compared between zones with the Mann–Whitney *U*-test. The sporozoite index (CSPI: the proportion of female anopheline mosquitoes carrying infective sporozoites in the head−thorax) for each visit was determined from mosquitoes from both the night catch and the spray catches. When many mosquitoes were caught, a sample of both catches was analysed; the size of the sample being in proportion to the size of the catch of each method.

3. Results

The results of the mosquito collections and analysis are shown in Table 1 for *A. gambiae s.l.* and Table 2 for *A. funestus*.

3.1. Abundance of vectors and vector composition

In both zones, *A. gambiae s.l.* and *A. funestus* were the only vectors of human malaria encountered. In the irrigated zone, *A. gambiae s.l.* represented 98–99% of the vectors (spray catch and night catch, all visits pooled). In the non-irrigated zone, 93% of the vectors caught in the spray catches were *A. gambiae s.l.* In the night catches in this zone, *A. funestus* was caught relatively more frequently, representing 20% of the vectors (all visits pooled). The relative abundance of the species varied over seasons and catching methods. In the villages of the irrigated zone, for all visits, *A. gambiae s.l.* was more abundant than *A. funestus* in the spray catches. In the night catch of October 1997 in Niessoumana and of January 1997 in Ténégué, more *A. funestus* were caught. In the villages of the non-irrigated zone, for the visits during the cold dry season (January 1997 and February 1998), *A. funestus* was more abundant than *A. gambiae s.l.* This was apparent both from the night catch and the spray catch. During the rainy season, the density of *A. gambiae s.l.* was higher than that of *A. funestus* in the irrigated zone.

Within the *A. gambiae s.l.* complex, *A. gambiae s.s.* was predominant over *A. arabiensis* in both zones. In the irrigated zone and in the non-irrigated zone, 99.6 and 96%, respectively, of the tested *A. gambiae s.l.* were *A. gambiae s.s.* Of *A. gambiae s.s.*, the Mopti chromosomic form was dominant (98.6 and 98.2% for the irrigated zone and the non-irrigated zone, respectively). The highest percentage of the Savana form was found during the visit of September 1995, and was 4.2 and 5.4% for the irrigated zone and the non-irrigated zone, respectively.

Table 1
Entomological parameters of *A. gambiae* in three villages in the irrigated zone and in three villages in the non-irrigated zone

Village	Date	Density (SC)	AP	FS	*n* (AP, FS)	HBI	*n* (HBI)	Density (NC)	PR	*n* (PR)	CSPI (%)	*n* (SI)	EIR (×0.01)
Irrigated													
Niessoumana	26 August 1995	230.51	0.64	0.88	24420	0.1	250	574.9	0.4	100	0.04	2513	0.5
	05 April 1996	176.67	0.66	0.69	13851	0.2	208	408.4	0.8	142	0	2152	0
	04 September 1996	57.26	0.56	0.64	3182	0.56	240	359	0.59	150	0.15	2062	1.6
	22 January 1997	3.18	0.55	0.92	172	0.75	119	16.1	0.86	80	1.53	391	1.8
	13 April 1997	130.26	0.5	0.78	6784	0.33	170	506.8	0.48	113	0.12	2600	1.9
	17 October 1997	3.39	0.52	0.71	126	0.69	68	14.3	0.85	55	0.97	309	0.8
	11 February 1998	3.97	0.55	0.83	303	0.63	180	11.3	0.9	61	2.18	641	2.5
Ténégué	14 August 1995	541.11	0.57	0.92	24604	0.25	332	568.6	0.62	50	0.09	2142	6.6
	09 April 1996	42.09	0.53	0.58	2854	0.33	499	322.8	0.88	151	0	2230	0
	18 August 1996	94.5	0.71	0.71	8371	0.26	240	489	0.53	150	0.05	2051	0.6
	06 January 1997	0.84	0.51	0.86	43	0.32	34	2.1	0.76	17	6	100	0.7
	17 April 1997	110.04	0.56	0.84	6042	0.32	180	225.9	0.52	114	0.04	2330	0.7
	06 October 1997	2.81	0.65	0.32	99	0.68	19	40.8	0.86	122	0.25	396	0.1
	07 February 1998	0.5	0.58	0.91	45	0.85	34	0.9	1	6	1.18	85	0.3
Tissana	19 August 1995	265.02	0.62	0.94	17127	0.17	224	552.3	0.45	100	0.33	2126	8.4
	31 March 1996	10.77	0.6	0.81	1168	0.48	497	25.5	0.84	108	0.34	2084	0.8
	24 August 1996	120.57	0.62	0.84	8688	0.26	240	230.1	0.62	148	0	2049	0
	10 January 1997	0.42	0.66	0.82	61	0.69	49	2	0.94	16	2.78	108	0.4
	22 April 1997	101.54	0.57	0.93	5448	0.43	120	54.5	0.64	103	0.1	2020	2.3
	09 October 1997	5.04	0.5	0.71	200	0.63	125	28.5	0.88	89	0.71	567	0.8
	03 February 1998	0.22	0.33	0.92	12	0.88	8	1.4	0.9	10	4.44	45	0.3
Non-irrigated													
Dokobougou	19 August 1995	20.35	0.56	0.94	2665	0.64	102	50.9	0.73	75	0.4	1508	2.7
	15 April 1996	2.75	0.46	0.87	256	0.66	150	3.5	0.95	20	0	580	0
	28 August 1996	37.64	0.62	0.81	4700	0.67	239	110.1	0.77	150	0.38	1042	4.8
	13 January 1997	0.05	0.1	1	1	1	1	0.3	1	2	0	12	0
	04 April 1997	3.5	0.61	0.86	462	0.68	104	1.3	0.7	10	0	557	0
	11 October 1997	24.82	0.44	0.87	1244	0.72	240	178.7	0.9	120	1.56	1606	10.6
	19 February 1998	0.04	0.75	1	6	0.67	3	0		0	0	8	0
Kalanampala	29 August 1995	22.6	0.44	0.99	2031	0.31	115	26.1	0.86	100	0.99	1813	3.1
	26 March 1996	0			0		0	0		0		0	0
	07 September 1996	31.45	0.52	0.78	3298	0.75	240	32.8	0.76	133	1.39	1009	13.3
	25 January 1997	0.01	1	0.5	2	1	1	0		0	0	3	0
	26 April 1997	1.99	0.29	0.94	124	0.82	67	0.3	1	1	0	427	0
	20 October 1997	4.8	0.51	0.99	293	0.78	180	4.2	0.94	18	5.71	420	10.7
	23 February 1998	0			0		0	0		0		0	0
Toumakoro	23 August 1995	14.7	0.56	0.96	1586	0.65	126	32.6	0.78	89	0.69	1602	3.5
	19 April 1996	3.58	0.51	0.88	353	0.76	223	1	1	3	0.29	693	0.4
	30 August 1996	15.89	0.51	0.95	1603	0.58	248	25.6	0.91	137	1.23	1055	5.5
	17 January 1997	0.03	0.33	1	2	1	2	0		0	0	6	0
	09 April 1997	5.73	0.62	0.67	604	0.58	104	3.8	0.64	22	0	990	0
	14 October 1997	14.04	0.48	0.91	754	0.79	201	36	0.92	95	1.41	1064	6.8
	15 February 1998	0	1	1	1	1	2	0		0	0	1	0

Density: mosquitoes per person per night; SC: spray catch. AP: active proportion (proportion of females that presumably tried to feed the night before in the SC (unfed, fed and semi-gravid); FS: feeding success; *n*: number of mosquitoes tested; HBI: human blood index (antropophily); NC: night catch; PR: parity ratio; CSPI: sporozoite index. EIR: entomological inoculation rate.

Table 2
Entomological parameters of *A. funestus* in three villages in the irrigated zone and in three villages in the non-irrigated zone

Village	Date	Density (SC)	AP	FS	*n* (AP, FS)	HBI	*n* (HBI)	Density (NC)	PR	*n* (PR)	CSPI (%)	*n* (SI)	EIR (×0.01)
Irrigated													
Niessoumana	26 August 1995	0.81	0		0	0.17	83	1.5		0	0	145	0
	05 April 1996	2.31	0.58	0.85	158	0.35	124	11.6		0	0	365	0
	04 September 1996	1.2	0.64	0.61	77	0.68	37	0.8		0	0	131	0
	22 January 1997	0.82	0.38	0.87	31	0.67	27	5.4	0.93	27	1.83	109	0.3
	13 April 1997	0.57	0.8	0.81	48	0.33	24	5.8	0.64	22	0	99	0
	17 October 1997	1.86	0.33	0.7	44	0.68	25	16.2	0.93	55	0	231	0
	11 February 1998	0.12	0.29	1	5	0.6	5	1	1	6	4	25	0.1
Ténégué	14 August 1995	0.59	0		0	0.32	28	0.8		0	0	56	0
	09 April 1996	0.6	0.58	0.82	45	0.44	36	8.8		0	0	148	0
	18 August 1996	0.13	0.75	1	12	0.43	7	0		0	0	16	0
	06 January 1997	0.39	0.44	0.82	17	0.56	9	4.4	0.91	34	1.35	74	0.1
	17 April 1997	2.55	0.49	0.92	123	0.37	57	8.5	0.8	35	0	252	0
	06 October 1997	0.31	0.35	0.67	6	0.75	4	7.3	0.97	30	1.56	64	0.1
	07 February 1998	0.05	0.75	1	6	0.88	8	0.3	1	2	7.69	13	0.3
Tissana	19 August 1995	0.4	0		0	0.29	21	0		0	0	42	0
	31 March 1996	0.87	0.53	0.93	83	0.44	61	2.5		0	0	178	0
	24 August 1996	0.25	0.59	0.88	17	0.25	8	1		0	0	36	0
	10 January 1997	0.05	0.45	0.8	5	0.75	4	0.4	0.67	3	7.14	14	0.1
	22 April 1997	1.47	0.65	0.99	90	0.4	60	2.4	0.67	15	0.6	168	0.2
	09 October 1997	0.87	0.49	0.76	34	0.77	22	6.5	0.96	23	0.93	108	0.2
	03 February 1998	0.05	0.67	1	6	1	5	0.1		0	0	10	0
Non-irrigated													
Dokobougou	19 August 1995	0.44	0		0	0.83	77	0.1		0	0	101	0
	15 April 1996	1.28	0.48	0.98	125	0.38	120	1.9	1	13	0	275	0
	28 August 1996	0.86	0.5	0.86	87	0.75	44	1.8		0	0.58	171	0.2
	13 January 1997	1.43	0.31	0.96	94	0.49	76	1.3	0.89	9	0.32	308	0.1
	04 April 1997	0.87	0.67	0.98	127	0.45	76	1.5	0.91	11	0	196	0
	11 October 1997	3.05	0.49	0.94	172	0.8	90	103.3	0.96	74	0.66	457	0.7
	19 February 1998	0.53	0.6	0.83	66	0.73	49	0.6	1	5	0	15	0
Kalanampala	29 August 1995	0.11	0		0	0.55	11	0		0	0	23	0
	26 March 1996	0.02	0.67	0.5	2		0	0		0	0	3	0
	07 September 1996	0.15	0.45	0.93	14	0.64	11	0.1		0	0	31	0
	25 January 1997	0.02	0		0		0	0		0	0	3	0
	26 April 1997	0.02	0.6	1	3		0	0		0	0	4	0
	20 October 1997	0.05	0.17	1	1	1	2	0		0	0	6	0
	23 February 1998	0			0		0	0		0		0	0
Toumakoro	23 August 1995	0.23	0		0	0.81	31	0.4		0	0	48	0
	19 April 1996	0.47	0.62	0.96	56	0.63	43	0.6	1	2	0	94	0
	30 August 1996	0.36	0.6	0.98	43	0.62	29	1.3		0	0	55	0
	17 January 1997	0.23	0.46	0.79	19	1	15	0.6	1	5	0	45	0
	09 April 1997	0.37	0.68	0.86	42	0.69	39	1	1	8	0	64	0
	14 October 1997	4.28	0.47	0.99	226	0.82	93	15.7	0.92	59	0.51	589	0.8
	15 February 1998	0.41	0.45	0.92	38	0.82	33	0		0	0	85	0

Legend: see Table 1.

In general, the density of *A. gambiae s.l.* was a factor 10 or more higher in the irrigated zone than in the non-irrigated zone. However, during the visit of October 1997 (end of the rainy season), the density of *A. gambiae s.l.* was higher in the villages Dokobougou and Toumakoro (in the non-irrigated zone) than in the villages in the irrigated zone. The densities of *A. funestus* were similar in both zones. An exception to this was the village of Kalanampala, which showed very low densities of *A. funestus* during all visits.

3.2. Comparison of night catch and spray catch

In the irrigated zone, the ratio between the estimates of the number of bites per person per night from the night catch (NC) and the spray catch (SC) (corrected for anthropophily and feeding status) showed no significant correlation with mosquito density (as estimated by the geometric mean of both methods) for both *A. gambiae s.l.* ($r = 0.04$) and *A. funestus* ($r = 0.1$). The relationship between the two methods could therefore be described as linear. With a linear regression analysis of the estimates of the night catch against those of the spray catch (abscissa forced to 0), ratios of 12.5 (95% confidence interval (CI): 8.0–17.1; $r = 0.52$) and 23.4 (95% CI: 15.0–31.7; $r = 0.55$) were found for *A. gambiae s.l.* and *A. funestus*, respectively. However, for the non-irrigated zone, the ratio between the estimates of the number of bites per person per night from the night catch and the spray catch did show a significant correlation with mosquito density for both *A. gambiae s.l.* ($r = 0.76$) and *A. funestus* ($r = 0.92$). With a linear regression analysis of the logarithmically transformed estimates of the night catch against those of the spray catch, the following relationship for *A. gambiae s.l.* was found:

- $NC = 1.73\ (SC)^{1.77}$ ($P < 0.0001$, $r = 0.93$).

For *A. funestus* this was:

- $NC = 17.2\ (SC)^{1.56}$ ($P < 0.0001$, $r = 0.92$).

3.3. Parity ratio, anthropophily and feeding success

For *A. gambiae s.l.*, the parity ratio (the proportion of female mosquitoes having laid eggs at least once) was negatively correlated to mosquito density (estimated by the night catch method). In a regression analysis with the parity ratio transformed using the logit function (to make it a continuous variable), and the density logarithmically transformed, this relationship was significant for both the irrigated zone and the non-irrigated zone ($r = -0.77$, $P < 0.0001$ and $r = -0.68$, $P < 0.01$ for the irrigated and non-irrigated zone, respectively). For all visits, the parity ratio of *A. gambiae s.l.* was higher in the non-irrigated zone than in the irrigated zone.

The anthropophily (as measured by the proportion of the blood meals of human origin) was positively correlated with the parity ratio. This was significant for *A. gambiae s.l.* in the irrigated zone ($r = 0.73$, $P < 0.001$) and in the non-irrigated zone ($r = 0.64$, $P < 0.01$), and for *A. funestus* in the irrigated zone ($r = 0.64$, $P < 0.05$). The anthropophily was negatively correlated with the active mosquito density (estimated by the spray catch). The strongest correlations were found with the densities of *A. gambiae* and *A. funestus* pooled. This relationship was significant for *A. gambiae s.l.* and for *A. funestus* for the irrigated zone ($r = -0.86$, $P < 0.0001$), but not for the non-irrigated zone (the proportion of bites on humans was transformed using the logit function, and the density was logarithmically transformed). There was no significant correlation between the feeding success and the total biting density (*A. gambiae s.l.* and *A. funestus* combined), between the feeding success and the anthropophily or between the feeding success and the parity ratio. The average feeding success in the irrigated zone was 0.84 and 0.85 for *A. gambiae s.l.* and *A. funestus*, respectively. In the non-irrigated zone this was 0.88 and 0.94. For both species, the feeding success was significantly higher in the non-irrigated zone than in the irrigated zone (odds ratio = 1.64; 95% CI: 1.56–1.75 and odds ratio = 3.44; 95% CI: 2.56–5.26, for *A. gambiae s.l.* and *A. funestus*, respectively, Mantel–Haenszel χ^2-test stratified by survey).

3.4. Sporozoite index

In the irrigated zone, the sporozoite index was highest for *A. gambiae s.l.* and *A. funestus* during the cold dry season (visits of January 1997 and February 1998), when the mosquito densities were low. In this season in the non-irrigated zone, however, only one

sporozoite-positive *A. funestus* was detected. During the rainy season (visits of August/September and October), the sporozoite index was significantly higher for *A. gambiae s.l.* in the non-irrigated zone (mean = 0.0153) than in the irrigated zone (mean = 0.0029) for all visits (Mann–Whitney *U*-test, $P < 0.05$). In general, both for *A. gambiae s.l.* and *A. funestus*, the sporozoite index was higher in October (end of the rainy season) (mean of *A. gambiae s.l.* = 0.0098, mean of *A. funestus* = 0.0073) than in August/September (mean of *A. gambiae s.l.* = 0.0052, mean of *A. funestus* = 0.0005).

3.5. Entomological inoculation rate (EIR)

The transmission in both zones was mainly due to *A. gambiae s.l.* During the cold dry season, when transmission was low, transmission by *A. funestus* gained at bit in importance. The combined transmission by *A. gambiae s.l.* and *A. funestus*, as estimated by the spray catch, is shown in Fig. 2. As these figures are not corrected for exophilic behaviour, absolute values of these estimates should be used with care. In the villages in the non-irrigated zone, the EIR was mostly below detection level during the dry season. At the end of the rainy season, however, the EIR rose rapidly to a high level of over 0.075 infectious bites per person per night. In the villages in the irrigated zone, the transmission was fairly constant over the seasons and at a low level between 0 and 0.025 infectious bites per person per night. However, in Ténégué and Tissana, the transmission by *A. gambiae s.l.* was at an elevated level in August 1995 (EIR of 0.066 and 0.083, respectively), whereas it was very low in August 1996 (EIR of 0.006 and 0, respectively). The results of the Mann–Whitney *U*-test ($P < 0.05$) showed that during the cold dry season, the EIRs were significantly higher in the villages in the irrigated zone (mean = 0.0115) than in those of the non-irrigated zone (mean = 0.0002). For the hot dry season, this difference was less clear: the visit of April 1996 did not show significant difference, and the visit of April 1997 showed higher EIRs in villages in the irrigated zone. For both visits combined, the EIR was significantly higher in the irrigated zone (mean = 0.0098) than in the non-irrigated zone (mean = 0.0007). For the middle of the wet season, the visit of August 1995 did not show a significant difference between the zones. In the visit of August 1996, however, the EIRs were significantly higher in the villages in the non-irrigated zone (mean = 0.0793) than those in the irrigated zone (mean = 0.0073). For both visits combined, this difference was not significant. At the end of the rainy season (October 1997), the EIRs were higher in the villages in the non-irrigated zone (mean = 0.0987) than those in the irrigated zone (mean = 0.0067).

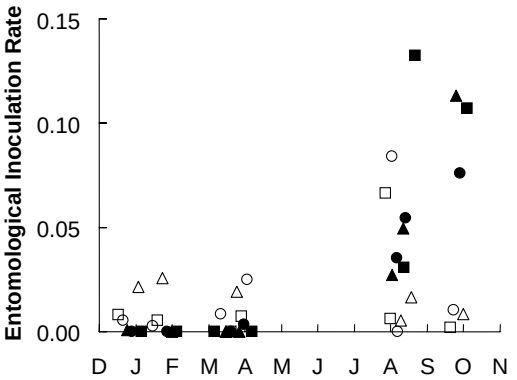


Fig. 2. The entomological inoculation rate (infectious bites per person per night) of *A. gambiae s.l.* and *A. funestus* combined. Inoculation rates are plotted on the time scale of 1 year to show seasonal variation. The entomological inoculation rates (EIRs) of the three villages in the irrigated zone are indicated with open symbols ((△) Niessoumana, (□) Ténégué, (○) Tissana), and those of the three villages in the non-irrigated zone with solid symbols ((▲) Dokobougou, (■) Kalanampala, (●) Toumakoro). The number of bites was calculated by multiplying the human blood index by the number of blood-fed and semi-gravid female mosquitoes per sleeper (collected by the spray catch).

4. Discussion

In both zones, the vector species composition was very similar. *Anopheles gambiae s.l.* consisted mainly of the Mopti form of *A. gambiae s.s.*, both in the rainy and dry season. This is typical for the irrigated/flooded areas of North and East Mali, as observed by Touré et al. (1998a). In a study comparing rice-cultivating villages with non-rice cultivating villages in Burkina Faso, Robert et al. (1989) also almost exclusively observed *A. gambiae s.s.* of the Mopti form in the rice cultivating villages, but found that *A. gambiae s.s.* Savanna form was mostly dominant in the non-rice growing villages, especially those in dryer areas. *Anopheles arabiensis* was also

found in that study. Both member-species of the *A. gambiae* species complex seem to compete for the same niche in sunlit irrigated fields (until the growing crop screens the larval breeding places from solar radiation (Chandler and Highton, 1975; Robert et al., 1988a). Where the Mopti form of *A. gambiae s.s.* is absent (e.g. Senegal, Burundi, Cameroon and Kenya), *A. arabiensis* can maintain itself in habitats under permanent irrigation (Robert et al., 1992; Petrarca et al., 1987; Ijumba et al., 1990). However, Lindsay et al. (1998) found that *A. gambiae s.s.* dominated in saturated environments, whereas *A. arabiensis* was more common in areas subject to desiccation. This is not the case in the non-irrigated zone in Mali.

The density of *A. gambiae s.l.* mosquitoes was generally higher in the irrigated zone, except at the end of the rainy season. Many other studies have observed higher densities of mosquitoes in rice-cultivating areas as compared to neighbouring areas without irrigated rice cultivation (Coosemans, 1985; Robert et al., 1985; Faye et al., 1993, 1995; Dossou-Yovo et al., 1994; Marrama et al., 1995; Chandler et al., 1975; Ijumba et al., 2002). The abundance of *A. funestus* in the non-rice growing villages of Dokobougou and Toumakoro was comparable to its abundance in the villages of the irrigated zone. Kalanampala, however, had a very low *A. funestus* population. Although Dokobougou and Toumakoro were 10–15 km away from the nearest rice fields, one cannot exclude possible dispersion over this distance. More mosquitoes were present in these villages in April 1996 than normally observed during the preceding months, despite the absence of rain. These mosquitoes could have been dispersed from the irrigated fields, in which, at that time of the year, the off-season crop was being cultivated. However, for *A. gambiae*, dispersal from breeding sites over such distances is not commonly observed. A mark-release-recapture study in an area near Bamako, Mali, showed that *A. gambiae s.l.* generally does not disperse further than 1 km (Dolo, 1996). In Senegal, Faye et al. (1993) observed a drop in biting density for *A. gambiae s.l.* from 17 bites per person per night at the border of the rice field to 1 bite per person per night at 5 km distance. Kalanampala, which is situated at 17 km distance of the closest rice field, did not show any unexpected presence of mosquitoes during the dry season. It is possible that a few migrating mosquitoes from the irrigated zone enhanced the initial build-up of the population in the non-irrigated zone at the start of the rainy season.

In the irrigated zone, the night catch method estimated the number of bites per person per night a factor 12 higher than the spray catch method, independent of the mosquito biting density. Pull and Grab (1974) found only a factor 1.8 more bites in the night catch than in the spray catch in Kenya. This difference could be due to the presumed lack of mosquito nets during the study period (1972−1973) in Kenya, but also to differences in sampling methodology (only indoor sampling in the Kenya study). People sitting unprotected might be an easy target for mosquitoes, and therefore receive proportionally more bites than they would have in a normal protected situation. Furthermore, in a situation where a large proportion of the population is using mosquito nets, the nets will divert the mosquitoes towards the capturers. The fact that, in the irrigated zone, the ratio between the estimates of the number of bites was constant with mosquito density, agrees with the observation that in these villages, everybody was sleeping under a (not impregnated) mosquito net year round. In the non-irrigated zone, however, this ratio increased with mosquito density. A possible explanation for this is that in this zone, people only used mosquito nets during the rainy season with high mosquito densities. In the dry season, nuisance was very low, and the mosquito nets were not used, as the mosquito nets in the non-irrigated zone were uncomfortable—they were made of densely woven nylon or cotton fabric, with little or no ventilation. However, as only about 20% of the people (mostly older men) owned mosquito nets in the non-irrigated villages, it is not likely that this percentage could have caused the positive correlation of the ratio between the estimates by the two methods of the number of bites with mosquito density. The lack of correlation of the proportion of bites on humans and of feeding success with active density supports this. As the night catch was most likely influenced by differential use of mosquito nets, the spray catch method was preferable for a comparison of the transmission of malaria between the two zones. Similar observations were made in other studies (Robert et al., 1985; Pull and Grab, 1974; Githeko et al., 1993b).

The parity ratio is often used as a parameter for the estimation of the mosquito longevity, together with the length of the first gonotrophic cycle. However, as dif-

ferences in recruitment rate can have a major impact on the parity ratio, it is necessary to correct for this if the mosquito population is not relatively stable (Briët, 2002). In this study, it is clear that there were large fluctuations in mosquito density over the seasons, and that, therefore, there would be a large error of recruitment in such calculation. Another possibility is to take the average parity ratio over a whole year, but this rules out the estimation of seasonal or density-dependent adult mortality. In this study, large intervals between observations and large differences for the same period in different years compromise the calculation of the average parity ratio over the period of a year. However, the fact that the parity ratio of *A. gambiae s.l.* was generally lower in the irrigated zone indicates that the daily mortality rate might be higher in the irrigated zone than in the non-irrigated zone, assuming that the length of the first gonotrophic cycle would be similar in both zones. Robert et al. (1985) made a similar observation in Burkina Faso, where they found a difference in average parity ratios between a village in an irrigated area and a village with no irrigation (0.46 and 0.61, respectively). These authors also found a lower anthropophily in the mosquitoes in the village in the irrigated area. Various other studies found low average parity ratios in villages with rice cultivation (Tia et al., 1992; Charlwood et al., 1997), sometimes in combination with a low anthropophily (Robert et al., 1992; Lindsay et al., 1991), or lower parity ratios as compared to villages in neighbouring areas without rice cultivation (Faye et al., 1995; Dossou-Yovo et al., 1994). As the density of mosquitoes is generally much higher in the irrigated zone than in the non-irrigated zone, it is possible that competition (mechanisms could include stress for early reproduction in rapidly growing populations), and increased chance of diseases at higher densities might be responsible for a difference in mortality rates and ensuing parity ratios. Unfortunately, in our study, no data on mosquito densities of non-vector species were collected, for a better estimation of the effect of interspecific competition. The small but statistically significant difference in feeding success may also be caused by differential use of (non-impregnated) mosquito nets, with an impact on mortality. Trials have shown impregnated mosquito nets to have negative impact on anthropophily, longevity or both (Magesa et al., 1991; Magbity et al., 1997; Carnevale et al., 1988).

There were large seasonal differences observed in the anthropophily of both *A. gambiae s.l.* and *A. funestus*. As *A. gambiae s.l.* consisted mainly of the Mopti form of *A. gambiae s.s.*, with little variation over the seasons, it is not very likely that such differences were caused by succession of different subspecies with different host preferences (Lindsay et al., 1991). Moreover, Touré et al. (1998a) did not find such differences in anthropophily and endophily between three reproductive units within *A. gambiae s.s.* in Mali. Presence of cattle can have a major influence on the host choice of mosquitoes (Garrett-Jones et al., 1980), much more so than pigs and sheep (Robert et al., 1988b). In the irrigated zone, oxen used for traction, asses and some milking cows are present year round. Most families possessed one or two asses, and a couple of bovines. In Niessoumana, on average there was one large animal (bovine or ass) for every four humans, whereas this ratio was 1:2.5 for Ténégué and Tissana. From December to June, cattle, which are kept on pastures outside the irrigated perimeter during the rainy season, come into the fields to feed on straw left after harvest. As the anthropophily was low both in April and August, the migration of this cattle does not seem to have an effect. In the non-irrigated zone, where asses are also present year round, nomadic tribes (the Peulh) bring in their cattle to graze the fields during the rainy season. The Peulh then live in camps close to the villages. At the end of the rainy season, the Peulh move on to the south. Here also, anthropophily is low both in April (dry season) and August (rainy season). Only in one village (Kalanampala), the index was much lower in August, which could have been caused by the presence of the nomadic cattle. Competition at higher densities in the continuous presence of asses or milking cattle seems to have much more effect on seasonal fluctuations in anthropophily than migration of cattle. In contrast, Vercruysse (1985) mentions that seasonal movement of cattle influenced the anthropophily in Senegal. Especially in the irrigated zone, it is possible that due to competition at high densities, mosquitoes were diverted to livestock. It is striking that, in general, in the irrigated rice growing villages, anthropophily was much lower than in the villages without irrigated agriculture, for both *A. gambiae s.l.* and *A. funestus*. The lower proportion of bites on humans in areas with irrigated rice cultivation may be the result of the use of mosquito nets, because of the high mosquito

densities. A similar explanation was given by Robert and Killick Kendrick (as referred to in Robert et al. (1985), who observed a human blood index of 0.72 in a non-irrigated savannah zone (with a low mosquito density) and of 0.40 in an irrigated rice-cultivating area with a high mosquito density, and where all inhabitants slept under mosquito nets year round (Robert et al., 1991a). Githeko et al. (1993b) found lower anthropophily for *A. gambiae s.l.* in a rice-cultivating village as compared to a sugarcane-growing village (but did not give data on mosquito net use). Faye et al. (1993) found a lower anthropophily for *A. pharoensis* in an irrigated-rice-cultivating village as compared to a village with rain-fed agriculture in an area of Senegal, where mosquito net use is common. In studies where mosquito net use was compared between villages with different (annual) densities of mosquitoes, a larger proportion of the human population used nets in areas with a higher mosquito density (Thomson et al., 1994, 1996). An alternative explanation for the observed drop in anthropophily during the rainy season would be that young (nulliparous) mosquitoes could be more zoophilic than older (parous) mosquitoes. However, Robert et al. (1991b) showed the contrary in a study in Burkina Faso, as they found that those *A. gambiae* (mostly of the Mopti chromosomic form) that had fed on cattle had a higher parity ratio than those that had fed on humans.

For both *A. gambiae s.l.* and *A. funestus*, the observed seasonal changes in the sporozoite index in the irrigated zone correspond well to the seasonal changes in anthropophily and parity ratio. The chance of a mosquito picking up the human malaria parasite is positively correlated to its anthropophily. In an older mosquito population, a larger proportion of the mosquitoes is old enough to be able to harbour infective sporozoites, as the parasite needs an incubation period in the mosquito to mature (Lines et al., 1991b). Although seasonal or density-dependent adult mortality could not be shown is this study set-up, it is not unlikely that the observed low parity ratios at high mosquito densities were partially due to low mosquito longevities, which would have a major impact on infection rates, as a larger proportion of the mosquitoes would not live long enough to allow the maturation of the sporozoites. The fact that during the surveys at the peak (August 1995 and August 1996) and the end of the rainy season (October 1997), a higher sporozoite index was found for the non-irrigated villages than for the irrigated villages for *A. gambiae s.l.*, is similarly explained by the higher parity ratio and anthropophily of the mosquitoes in the non rice growing villages. For *A. funestus*, these differences were less clear.

The transmission was very low in the villages in the non-irrigated zone during the dry season. Although transmission was not detected in Kalanampala, it might not have been totally absent, as new cases of malaria were being recorded during this period (Sissoko et al., 2004). That in the non-irrigated zone, during the cold dry season, no infective *A. gambiae s.l.* were found, could be due to insufficient numbers of mosquitoes available for analysis to be able to detect anything. The high sporozoite index together with the high mosquito density during the rainy season causes a peak transmission during the wet season in the villages in the non-irrigated zone. In the villages in the irrigated zone, however, sporozoite indices are low when mosquito densities are high and vice versa, which results in a transmission that varies little over the seasons.

In conclusion, rice cultivation in the Sahel environment altered the transmission patterns from seasonal to perennial. However, annual transmission may have been reduced (Sissoko et al., 2004). In the non-irrigated zone, reduction of transmission might substantially reduce incidence (Sissoko et al., 2004). Such reduction could be achieved by bed net marketing programs with a strong extension component on the necessity of their use, even at low mosquito densities. In the irrigated zone, vector control should focus on reducing vector-human contact and mosquito longevity rather than larva control.

Acknowledgements

This study was undertaken within the framework of the WARDA/WHO–FAO–UNEP PEEM/IDRC/DANIDA/Norway Health Research Consortium on the Association between Irrigated Rice Ecosystems and Vector-borne Diseases in West Africa. The Consortium received financial support from the International Development Research Center (IDRC), Ottawa, Canada, the Danish International Development Agency (DANIDA), and the Royal Government of Norway. The authors wish to thank Dr. Doré Guindo

for providing information on cattle movements in the Office du Niger.

References

Briët, O.J., 2002. A simple method for calculating mosquito mortality rates, correcting for seasonal variations in recruitment. Med. Vet. Entomol. 16, 22–27.

Carnevale, P., Robert, V., Boudin, C., Halna, J.M., Pazart, L., Gazin, P., Richard, A., Mouchet, J., 1988. La lutte contre le paludisme par des moustiquaires imprégnées de pyréthrinoides au Burkina Faso. Bull. Soc. Pathol. Exot. Filiales 81, 832–846.

Chandler, J.A., Highton, R.B., 1975. The succession of mosquito species (Diptera, Culicidae) in rice fields in the Kisumu area of Kenya, and their possible control. Bull. Entomol. Res. 65, 295–302.

Chandler, J.A., Highton, R.B., Hill, M.N., 1975. Mosquitoes of the Kano plain, Kenya 1. Results of indoor collections in irrigated and nonirrigated areas using human bait and light traps. J. Med. Entomol. 12, 504–510.

Charlwood, J.D., Smith, T., Billingsley, P.F., Takken, W., Lyimo, E.O.K., Meuwissen, J.H.E.T., 1997. Survival and infection probabilities of anthropophagic anophelines from an area of high prevalence of *Plasmodium falciparum* in humans. Bull. Entomol. Res. 87, 445–453.

Charlwood, J.D., Smith, T., Kihonda, J., Heiz, B., Billingsley, P.F., Takken, W., 1995. Density independent feeding success of malaria vectors (Diptera: Culicidae) in Tanzania. Bull. Entomol. Res. 85, 29–35.

Coosemans, M.H., 1985. Comparison de l'endémie malarienne dans une zone de riziculture et dans une zone de culture de coton dans la plaine de la Rusizi, Burundi. Ann. Soc. Belg. Med. Trop. 65, 187–200.

Detinova, T.S., 1962. Age grouping methods in Diptera of medical importance, with special reference to some vectors of malaria. WHO Monograph, pp. 1–216.

Dolo, G., 1996. Etude des populations d' *An. gambiae s.l.* par marquage, lâcher et recapture à Banambani en 1993 et 1994 (Arrondissement central de Kati, Mali). Mémoire de DEA en Entomologie et Parasitologie médicales, ISFRA.

Dossou-Yovo, J., Doannio, J., Riviere, F., Duval, J., 1994. Rice cultivation and malaria transmission in Bouake city (Cote d'Ivoire). Acta Trop. 57, 91–94.

Faye, O., Fontenille, D., Gaye, O., Sy, N., Molez, J.F., Konate, L., Hebrard, G., Herve, J.P., Trouillet, J., Diallo, S., Mouchet, J., 1995. Paludisme et riziculture dans le delta du fleuve Sénégal. Ann. Soc. Belg. Med. Trop. 75, 179–189.

Faye, O., Fontenille, D., Herve, J.P., Diack, P.A., Diallo, S., Mouchet, J., 1993. Malaria in the sahelian zone of Senegal. 1. Entomological data concerning transmission. Ann. Soc. Belg. Med. Trop. 73, 21–30.

Garrett-Jones, C., Boreham, P.F.L., Pant, C.P., Jones, C.G., 1980. Feeding habits of anophelines (Diptera: Culicidae) in 1971–78, with reference to the human blood index: a review. Bull. Entomol. Res. 70, 165–185.

Githeko, A.K., Service, M.W., Mbogo, C.M., Atieli, F.K., Juma, F.O., 1993b. *Plasmodium falciparum* sporozoite and entomological inoculation rates at the Ahero rice irrigation scheme and the Miwani sugar-belt in western Kenya. Ann. Trop. Med. Parasitol. 87, 379–391.

Ijumba, J.N., Mosha, F.W., Lindsay, S.W., 2002. Malaria transmission risk variations derived from different agricultural practices in an irrigated area of northern Tanzania. Med. Vet. Entomol. 16, 28–38.

Ijumba, J.N., Mwangi, R.W., Beier, J.C., 1990. Malaria transmission potential of *Anopheles* mosquitoes in the Mwea–Tebere irrigation scheme, Kenya. Med. Vet. Entomol. 4, 425–432.

Klinkenberg, E., Takken, W., Huibers, F., Toure, Y.T., 2003. The phenology of malaria mosquitoes in irrigated rice fields in Mali. Acta Trop. 85, 71–82.

Lindsay, S.W., Parson, L., Thomas, C.J., 1998. Mapping the ranges and relative abundance of the two principal African malaria vectors, *Anopheles gambiae sensu stricto* and *An. Arabiensis*, using climate data. Proc. R. Soc. Lond. B Biol. Sci. 265, 847–854.

Lindsay, S.W., Wilkins, H.A., Zieler, H.A., Daly, R.J., Petrarca, V., Byass, P., 1991. Ability of *Anopheles gambiae* mosquitoes to transmit malaria during the dry and wet seasons in an area of irrigated rice cultivation in the Gambia. J. Trop. Med. Hyg. 94, 313–324.

Lines, J.D., Curtis, C.F., Wilkes, T.J., Njunwa, K.J., 1991a. Monitoring human-biting mosquitoes (Diptera: Culicidae) in Tanzania with light-traps hung beside mosquito nets. Bull. Entomol. Res. 81, 77–84.

Lines, J.D., Wilkes, T.J., Lyimo, E.O.K., 1991b. Human malaria infectiousness measured by age-specific sporozoite rates in *Anopheles gambiae* in Tanzania. Parasitology 102, 167–177.

Magbity, E.B., Marbiah, N.T., Maude, G., Curtis, C.F., Bradley, D.J., Greenwood, B.M., Petersen, E., 1997. Effects of community-wide use of lambdacyhalothrin-impregenated bednets on malaria vectors in rural Sierra Leone. Med. Vet. Entomol. 11, 79–86.

Magesa, S.M., Wilkes, T.J., Mnzava, A.E.P., Njunwa, K.J., Myamba, J., Kivuyo, M.D.P., Hill, N., Lines, J.D., Curtis, C.F., 1991. Trial of pyrethroid impregnated bednets in an area of Tanzania holoendemic for malaria. Part 2. Effects on the malaria vector population. Acta Trop. 49, 97–108.

Marrama, L., Rajaonarivelo, E., Laventure, S., Rabarison, P., 1995. *Anopheles funestus* and rice culture on the Plateau of Madagascar. Sante 5, 415–419.

Petrarca, V., Vercruysse, J., Coluzzi, M., 1987. Observations on the member species of the *Anopheles gambiae* complex in the Senegal River Basin. Med. Vet. Entomol. 1, 303–312.

Pull, J.H., Grab, B., 1974. A simple epidemiological model for evaluating the malaria inoculation rate and the risk of infection in infants. Bull. World Health Organ. 51, 507–516.

Robert, V., Gazin, P., Boudin, C., Molez, J.F., Ouedraogo, V., Carnevale, P., 1985. La transmission du paludisme en zone de savane arborée et en zone rizicole des environs de Bobo-Dioulasso (Burkina-Faso). Ann. Soc. Belg. Med. Trop. 65, 201–214.

Robert, V., Ouari, B., Ouédraogo, V., Carnevale, P., 1988a. La succession des espèces anophéliennes et le cycle du riz; étude écologique des *Culicidae* adultes et larvaires dans la rizière de la Vallée du Kou, Burkina Faso. Acta Trop. 4, 351–359.

Robert, V., Carnevale, P., Ouedraogo, V., Petrarca, V., Coluzzi, M., 1988b. Human malaria transmission in a savannah village of south-west Burkina Faso. Ann. Soc. Belg. Med. Trop. 68, 107–121.

Robert, V., Petrarca, V., Carnevale, P., Ovazza, L., Coluzzi, M., 1989. Analyse cytogénétique du complexe *Anopheles gambiae* dans la région de Bobo-Dioulasso, Burkina Faso. Ann. Parasitol. Hum. Comp. 64, 290–311.

Robert, V., Ouedraogo, V., Carnevale, P., 1991a. The transmission of human malaria in a village at the centre of the Kou Valley rice-growing area, Burkina Faso. In: Robert V. (Eds.), Le paludisme en Afrique de l'ouest: etudes entomologiques et epidemiologiques en zone rizicole et en milieu urbain. ORSTOM, Paris, pp. 5–15.

Robert, V., Petrarca, V., Coluzzi, M., Boudin, C., Carnevale, P., Chippaux, J. P., Diomande, L., 1991b. Etude des taux de parturite et d'infection du complexe *Anopheles gambiae* dans la riziere de la Vallee du Kou, Burkina Faso. In: Robert, V., Chippaux J.P., Diomande L. (Eds.), Le paludisme en Afrique de l'ouest: etudes entomologiques et epidemiologiques en zone rizicole et en milieu urbain. ORSTOM, Paris, pp. 17–35.

Robert, V., Van den Broek, A., Stevens, P., Slootweg, R., Petrarca, V., Coluzzi, M., Le Goff, G., Di Deco, M.A., Carnevale, P., 1992. Mosquitoes and malaria transmission in irrigated rice-fields in the Benoue valley of northern Cameroon. Acta Trop. 52, 201–204.

Sissoko, M.S., Dicko, A., Briët, O.J.T., Sissoko, M., Sagara, I., Keita, H.D., Sagoba, M., Rogier, C., Touré, Y.T., Doumbo, O.K., 2004. Malaria incidence in relation to rice cultivation in the irrigated Sahel of Mali. Acta Tropica (this issue).

Thomson, M.C., D'Alessandro, U., Bennett, S., Connor, S.J., Langerock, P., Jawara, M., Todd, J., Greenwood, B.M., 1994. Malaria prevalence is inversely related to vector density in The Gambia, West Africa. Trans. R. Soc. Trop. Med. Hyg. 88, 638–643.

Thomson, M., Connor, S.J., Bennett, S., D'Alessandro, U., Milligan, P., Aikins, M., Langerock, P., Jawara, M., Greenwood, B.M., 1996. Geographical perspectives on bednet use and malaria transmission in The Gambia, West Africa. Soc. Sci. Med. 43, 101–112.

Tia, E., Gazin, P., Molez, J.F., Kone, A., Lochouarn, L., 1992. Irrigated farming and mosquito nuisance. The example of Banzon (Burkina Faso). Sante 2, 114–118.

Touré, Y.T., Petrarca, V., Traoré, S.F., Coulibaly, A., Maiga, H.M., Sankaré, O., Sow, M., Di Deco, M.A., Coluzzi, M., 1998a. The distribution and inversion polymorphism of chromosomally recognized taxa of the *Anopheles gambiae* complex in Mali, West Africa. Parassitologia 40, 477–511.

Vercruysse, J., 1985. Etude entomologique sur la transmission du paludisme humain dans le bassin du fleuve Sénégal (Sénégal). Ann. Soc. Belg. Med. Trop. 65, 171–179.